Research Hypothesis

A bifurcated, channeled flow aircraft configuration with a centrally aligned rear-mounted turbofan inlet may improve aircraft-level aeropropulsive efficiency by modifying the upstream flow field and reducing aerodynamic losses associated with conventional enclosed airframe geometries.

Through optimization of cavity cross-sectional area, longitudinal curvature, and inlet transition geometry, the configuration may achieve favorable inlet total-pressure recovery and flow uniformity while maintaining acceptable aerodynamic drag, structural mass, bending stiffness, and torsional rigidity.

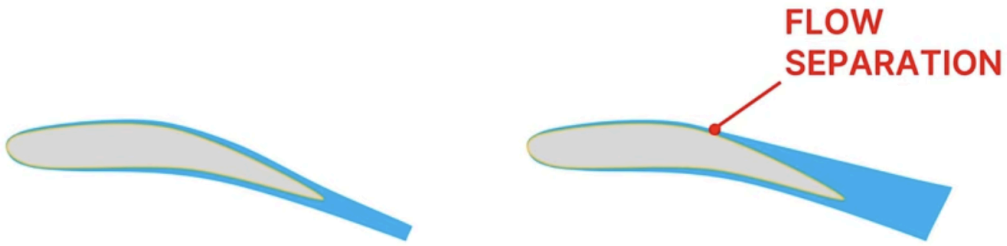
This investigation seeks to determine whether the aerodynamic and propulsive benefits of an integrated channeled flow configuration outweigh any additional wetted-area drag, flow interference, and structural penalties introduced by the cavity.

Key considerations in aircraft design

Frontal blockage: how much of the incoming airflow is obstructed by the aircraft's geometry, where projected frontal area is:

A_f = area of aircraft silhouette viewed from front

Reducing A_f can reduce pressure drag, especially when the removed geometry otherwise would cause substantial flow separation



However, a smaller frontal area doesn't automatically correlate to lower drag. A streamlined enclosed fuselage can have less drag than several thin, poorly arranged structural members

Wetted Surface Area: the total external surface area in contact with airflow, including contact surfaces that may intentionally allow air to pass through.

This becomes important because as air moves over a surface, viscosity creates a boundary layer and produces skin-friction drag simply defined as:

$$D_f = C_f \left(\frac{1}{2} \rho V^2\right) S_{wet}$$

Where

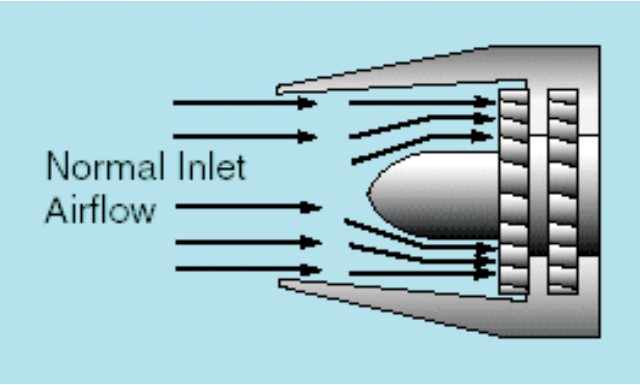
D_f = Skin-friction drag (N)
C_f = Skin-friction coefficient (dimensionless)
ρ = Air density (kg/m³)
V = Freestream velocity (m/s)
S_{wet} = Total wetted surface area (m²)

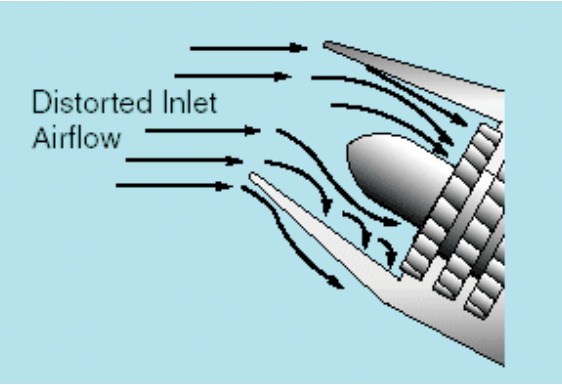
Interference Drag: additional drag caused by aerodynamic components interacting with one another, for example, the connection point between a wing and fuselage where airflow around each component then influences the other, which can then lead to:

Local pressure gradients
Boundary-layer thickening
Flow separation
Junction vortices

Inlet Pressure Recovery: the ratio of average total (stagnation) pressure at the exit of an engine inlet (compressor face) to the free-stream total pressure of the ambient air p_{2t}/p_{0t}, and a primary measure of how efficiently an aircraft or gas turbine inlet captures free-stream air and delivers it to the fan or compressor with minimal total pressure (energy) loss.

Distortion: non-uniform fluid flow, total pressure, total temperature, or swirl distribution reaching the face of the engine compressor or fan (at the Aerodynamic Interface Plane, or AIP)





Compressible-Flow Effects:

These occur when gas velocities approach or exceed the local speed of sound (typically defined at a Mach number $Ma > 0.3$), causing the gas density (ρ) to vary significantly alongside pressure and temperature.

Key effects include:

Variable Density and Continuity Coupling: Unlike incompressible flows where density is constant, compressible flow couples the mass conservation equation, momentum equations, and thermodynamics (the ideal gas law and energy equation). Changes in flow area or pressure directly alter gas density and velocity simultaneously.

Sonic Choking: In convergent-divergent nozzles, stator passages, or rotor blade throats, when the local flow velocity reaches the speed of sound ($Ma = 1$), the mass flow rate through the channel reaches a maximum limit for a given upstream condition. The turbine stage becomes "choked," meaning further reductions in downstream pressure cannot increase the mass flow rate.

Local Supersonic Regions and Shock Waves: As gas accelerates over the curved surface of a turbine blade (especially on the suction side), the local flow can easily exceed $Ma = 1$ even if the inlet or average velocity is subsonic ($Ma > 0.7$). This acceleration leads to the formation of local supersonic patches followed by normal or oblique shock waves.

Shock-Boundary Layer Interaction: When shock waves interact with the boundary layers along the turbine blade surfaces, they can induce flow separation, increase profile drag, and cause severe aerodynamic losses and high-cycle fatigue.

Prandtl-Meyer Expansion Fans: As high-speed gas rounds sharp trailing edges or concave/convex surface curves in turbine cascades, expansion fans form, rapidly accelerating the flow and dropping static pressure and temperature while converting internal energy into kinetic energy.

Aerodynamic Heating and Energy Conversion: Compressible flow is inherently tied to high energy regimes where kinetic energy converts into internal thermal energy (and vice versa), causing substantial temperature gradients across the blade rows.

Each of these interactions and considerations can increase drag beyond what the components would individually experience, which is a significant trade-off being evaluated in this study.

Selecting Parameters & Initial Calculations

Lift Coefficient and Mach Number:

In selecting a lift coefficient relevant to fast subsonic flight for comparison with commercial airlines and private jets, I'm selecting an assumed Mach number of 0.85 and a target lift coefficient (C_L) of 0.40 to allow for a balanced lift-to-weight ratio in the transonic regime (speed range in which flight speeds of air flowing around aircraft is concurrently subsonic, sonic, and supersonic). As well, in this speed range, aircrafts begin to experience issues with higher lift coefficients in which dynamic pressure is very high and the wings would generate much more lift than the aircraft weight, therefore leading it to ascend quickly rather than staying level at equal speed. Additionally, high angle of attack at Mach 0.85 forces airflow over the airfoil to accelerate extremely fast, which can lead to supersonic pocketing that results in a shock wave since the aircraft itself may be flying at Mach <1 but the local airflow velocity over the top of the wing is Mach >1. In this way, the benefit of a low lift coefficient at Mach 0.85 is then that the airflow acceleration is minimized over the top of the wing which keeps local supersonic flow minimal and delays the formation of shock waves which then reduces potential wave drag.

As well, under the assumption that this flight comparison takes place in the subsonic regime, comparable with commercial airliners and jets, I'll be using a cruising altitude of h = 11,000m for my further calculations.

For calculations, I'm referencing two distinct aircrafts that fly at subsonic speeds and carry similar footprints to my design. I'll be using the Northrop Grumman RQ-4B Global Hawk UAV primarily for weight and single-engine configuration reference, and the Gulfstream G650 for cruise speed reference.

The Global Hawk has a maximum takeoff weight (MTOW) of roughly 15,000kg, whereas the G650 has an MTOW of just over 45,000kg. I plan for my aircraft to primarily be for light payload/UAV applications similar to the Global Hawk (also considering monojet design), but on a larger scale with higher speeds, similar to the G650 (as well similar wing configuration), so I'll be using 30,000kg as the mass (m) for my aircraft, as it meets right in the middle of the intentionally selected high-weight jet and low-weight UAV.

Below are specifications of both selected comparison aircrafts:

Specification	**Gulfstream G650**	**NG RQ-4B Global Hawk**
Classification	Ultra-long-range business jet	High-altitude, long-endurance (HALE) surveillance UAV
Configuration	Swept wing, conventional tail	High-aspect-ratio straight wing, conventional tail
Max takeoff mass	45,178 kg	14,628 kg
Wingspan	30.35 m	39.8 m
Length	30.40 m	14.5 m
Height	7.82 m	4.7 m
Cruise Mach	0.85 long-range; 0.90 high-speed	~ 0.48 at 11 km (speed-based estimate)
Max operating Mach	0.925	Not publicly specified
Max operating altitude	15,545 m	18,288 m
Propulsion	2x Rolls-Royce BR725 turbofans	1x Rolls-Royce AE 3007H turbofan
Payload	Passengers and baggage	Up to 1,360kg internal payload
Crew	Piloted	Unmanned

Preliminary Flight Condition Calculations & Wing Sizing

Firstly we need to calculate the dynamic pressure (q) for CFD simulations and further calculations:

$q = 1/2\rho V^2$

Where

q = dynamic pressure (Pa)

ρ = air density (kg/m^3)

V = aircraft velocity relative to air (m/s)

Utilizing the selected cruise altitude of 11,000m, the International Standard Atmosphere (ISA) gives;

$\boxed{\rho = 0.3639 kg/m^3}$

So now calculating cruise velocity for our selected Mach 0.85, where Mach number is defined as:

$M = \frac{v}{a}$

And since at 11,000m, the speed of sound is:

$\boxed{\text{a} = 295.1 \text{ m/s}}$

$\therefore$ V = Ma = (0.85)(295.1 m/s)

$\boxed{\text{V} = 250.8 \text{ m/s}}$

Finally calculating dynamic pressure, we can substitute the values in:

$q = \frac{1}{2}(0.3639 kg/m^3)(250.8)^2$

$\boxed{\text{q} = 11.444.7 \text{ PA}}$

Now for calculating wing reference area (S) at MTOW, first we need the aircraft weight from our assumed value:

W = mg = (30,000kg)(9.81m/s^2)

$\boxed{\text{W} = 294{,}300 \text{ N}}$

Important to note here as well is that at steady flight, lift (L) is equal to weight (W), which in this case is lift required at MTOW

Wing reference area at MTOW is then:

$S = \frac{W}{qC_L}$

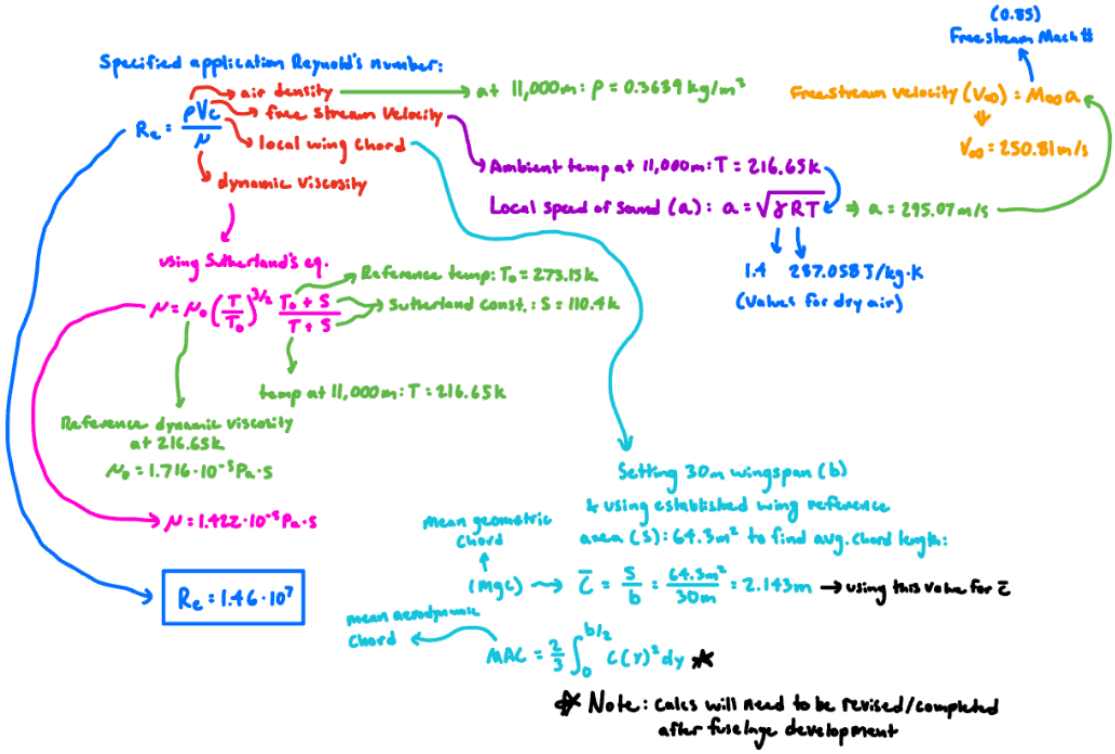
$S = \frac{294{,}300N}{(11{,}444.7Pa)(0.40)}$

$\boxed{S = 64.3m^2}$

Additional calculations for mass flow, freestream total pressure, and engine inlet flow to come

Initial material selection: aluminum (isotropic)

Calculation of Reynolds number using specified conditions:



Beyond initial calculations, the second phase of this investigation is aircraft optimization and airfoil selection. I decided to utilize XFOIL and XFLR5 for basic geometry screenings at a mach number of 0.6 to then take those passing candidates into further analysis in Ansys for further consideration. The parameters considered during this portion of the study are in the table below

XFOIL Airfoil Optimization Parameters:

Parameter	**Objective**	**Priority**	**Evaluation Method**
Lift coefficient (Cl)	Achieve required lift efficiently	Very High	Cl vs. angle of attack
Drag coefficient (Cd)	Minimize drag at target lift	Very High	Cd at Cl = 0.4

Lift-to-drag ratio (Cl/Cd)	Maximize aerodynamic efficiency	High	Cl/Cd across operating range
Pitching moment (Cm)	Minimize trim requirements	High	Cm at target Cl
Thickness-to-chord ratio (t/c)	Balance structural thickness and aerodynamic performance	Very High	Max thickness as % of chord
Pressure coefficient (Cp)	Evaluate suction peaks and pressure recovery	High	Upper/lower surface pressure distributions
Maximum lift coefficient (Cl, max)	Maintain low-speed lift capability	High	Max lift before stall
Lift-curve slope (dCl/dα)	Assess lift sensitivity to angle of attack	Medium	Slope of Cl vs. α

Study results to come